

360° Design thinking approach:(With reference to wallet)

1. Fact:

- a. Preferences:
 - i. Physical
 - ii. Material
 - iii. Weight
 - iv. Colour
 - v. Cost
 - vi. Brand

2. Form:

- a. Preferences
 - i. Shape
 - ii. Size
 - iii. Texture
 - iv. Pockets
 - v. Pattern

3. Function:

- a. What is the main use?
- b. Where/When do they use it?.
- c. Where it is stored when not in use?
- d. How is it carried?

4. Relationships:

- a. What gets added/removed daily from that product?
- b. What would he or she hate to lose in it?
- c. Is the product a reflection of their personality?
- d. How is it personalised?
- e. What does your partner have in the product which you don't have?

Explain the following terms with example:

Design:

Design is the purposeful and careful act of planning and organising elements to create or build something.

Example: Designing a Coffee Mug

Purpose and Functionality:

- *Design Objective:* Create a coffee mug that is comfortable to hold and keeps beverages hot.
- *Decisions:* Choose a handle shape that accommodates different hand sizes. Select materials that insulate well to maintain beverage temperature.

Innovation:

Innovation is a creative process where we apply our creativity to improve or modify the already existing solution according to customers needs

Example:

- Innovation: The introduction of smartphones revolutionized communication and changed the way people interact with information and technology.
- Features: Smartphones combine a phone, camera, computer, and various applications into a single device, offering versatility and convenience.

Invention:

It is the creation of something new that has not existed before directly from scratch. It is also a creative process where we apply our creative knowledge to build something new

Ex:

- Invention of aeroplane by wright brothers
- invention of car by karl benz

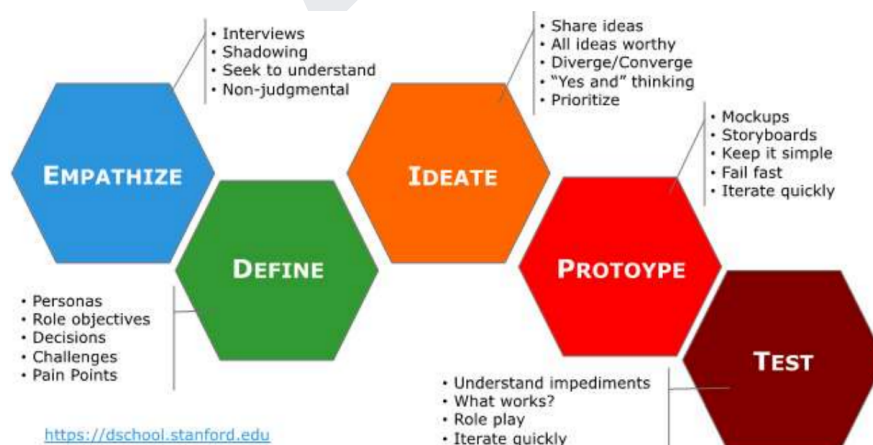
Creativity:

Creativity is a fundamental aspect of design thinking and is the ability to generate new ideas and solutions.

Ex:

Art: Painting, sculpture, photography, and other forms of art are all creative expressions. Artists use their imagination and skill to create works of art that are both beautiful and meaningful.

5 phases of design thinking approach:



- **Empathise:**
 - Understand what customers need by putting yourself in their shoes.
 - Gather information from all angles to see their challenges and desires.
- **Define:**
 - Clearly state why customers need a product or feature.
 - Describe their needs in specific and understandable terms.
- **Ideate:**
 - Develop new and creative ideas to meet customer needs.
 - Show these ideas as detailed drawings with sizes and shapes.
- **Prototype:**
 - Make a model of the product based on the drawings.
 - Create something tangible(touchable/real) to represent the planned product.
- **Test:**
 - Look at the final product from the customer's perspective.
 - Add or adjust details based on what the customer sees and feels.

Agile model of project management

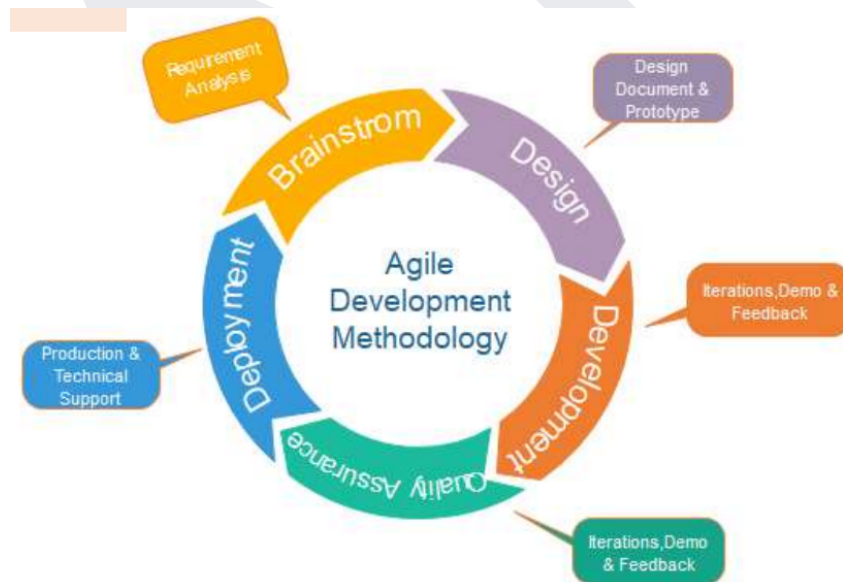
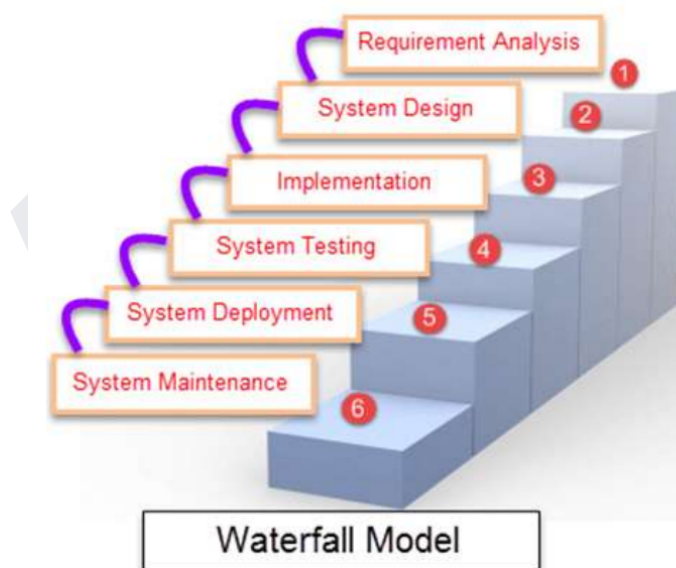


Fig. Agile Model

- **Brainstorm:**
 - Encourage open discussion among team members.
 - Collect diverse ideas and solutions.
- **Design:**
 - Create sketches, or mock-ups of the solution.

- Develop a clear plan for implementation.
- **Development:**
 - Break down tasks into smaller, achievable components.
 - Implement the solution in short, iterative cycles.
- **Quality Assurance:**
 - Conduct tests on individual components or features.
 - Address and fix any issues or bugs found..
- **Deployment:**
 - Release or deliver the solution to end-users.
 - Monitor and support the initial release for user feedback.

Waterfall model:



Requirement Analysis:

- Gather and document what the system needs to do.
- Understand and analyse user needs thoroughly.

System Design:

- Create a plan for how the system will work.
- Develop a detailed design for the system.

Implementation:

- Build the system according to the design.
- Write and create the software or solution.

System Testing:

- Test the complete system for issues and bugs.
- Address and fix any issues or bugs found.

System Deployment:

- Release the system to users or clients.
- Provide training if necessary.

System Maintenance:

- Support, update, and fix issues after deployment.
- Monitor and keep the system running smoothly.

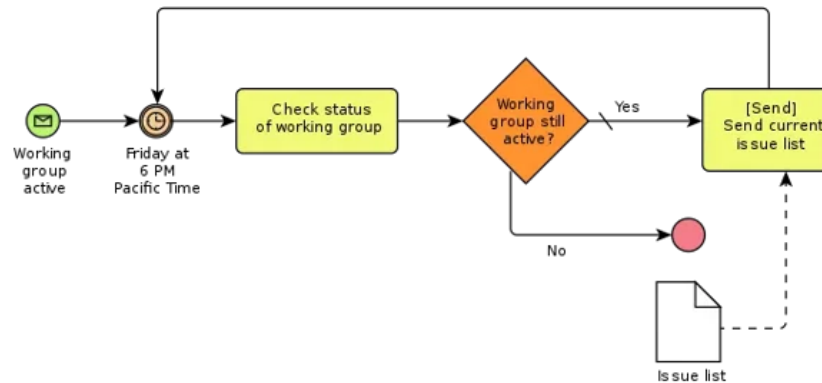
Compare agile and waterfall model:

	Agile	Waterfall
Approach:	Iterative and incremental approach	Sequential, linear approach where each phase is completed before moving to the next.
Flexibility:	Embraces changes and allows for continuous improvements at any phase of development.	Less adaptable to changes once a phase is completed.
Process Structure:	Phases are more flexible and can overlap. It consists of shorter cycles (sprints) for development.	Phases follow a strict sequence
Customer Involvement:	Continuous customer involvement and feedback throughout the development process.	Limited customer involvement during the development process, more focused on initial requirements gathering.

Advantage of agile model:

- ☐ **Team Collaboration:** Agile encourages teams to work together closely, leading to better communication and teamwork.
- ☐ **Flexibility:** Embraces changes and allows for continuous improvements at any phase of development.
- ☐ **Adaptability:** It allows for changes during the project, making it easier to respond to new requirements or issues.
- ☐ **Customer Involvement:** Continuous customer involvement and feedback throughout the development process.
- ☐ **Faster Results:** It delivers small parts of the project quickly, getting useful features to customers faster.

Note on Business process model(BPM):



- A business process model is a visual representation that shows how a business operates by breaking down its processes into manageable steps
- **They serve several critical purposes:**
 - **Communication and Understanding:**
 - Business process models act as a medium for communication between various stakeholders.
 - **Shared Understanding and Collaboration:**
 - These models promote shared understanding so that all involved parties can contribute their knowledge, insights, and improvements to the process.
 - **Combination of Process-related Steps:**
 - BPM involves various steps like process mapping, discovery, simulation, analysis, and improvement.
 - **Evolution of Efficiency Techniques:**
 - BPM, have evolved over time and have replaced previous approaches like Time and Motion Study (TMS) and Total Quality Management (TQM).

Advantages of BPM:

- **Clear Understanding:**
 - BPM shows how a business works in a clear picture, making it easy for everyone to understand.
- **Better Communication:**
 - It helps people talk and work together better by showing how things are done.
- **Better Decision-Making:**
 - It helps in making better choices by showing how things connect in the business.
- **Saving Money and Resources:**
 - It helps save money by finding and fixing wasteful or unnecessary steps in a process.
- **Finding Ways to Improve:**
 - BPM helps find better ways to do things by breaking processes into steps and making them easier to improve.

5 phases of design thinking in BPM:

- **Identifying the Challenge (BPM: Process Analysis):**
 - Recognizing an inefficient or problematic process within the organization.
- **Learning and Gathering Information (BPM: Process Understanding):**
 - Understanding why the process is inefficient by gathering relevant data and insights.
- **Identifying Opportunities (BPM: Solution Generation):**
 - Coming up with ways to make things better using the collected info.
- **Developing Ideas (BPM: Solution Implementation):**
 - Making detailed plans to fix the identified problems.
- **Testing and Evolving (BPM: Continuous Improvement):**
 - Testing new solutions and making them better based on feedback.

Agile in virtual collaboration

1. Allow Transparency:

- Transparency is one of the core principles of Agile.
- Do not keep responsibilities a secret and be clear over job functions.

2. Establish a Culture of Continuous Improvement:

- Drive improvements within the team.
- Recognize potential improvement opportunities and obtain ideas from the team.

3. Communication:

- A strong emphasis on communication is the key
- Communication is perhaps the most important artefact in any team and how you structure your communication will determine the output.

4. Construction of a Rhythm:

- There are three things which are absolute necessary in agile methodology:
 - Transparency
 - inspection
 - adaptation.
- Scrum also calls it the three pillars of Scrum

5. Generate a Culture of Courage & Flexibility:

- Establish an environment where the team is courageous to take steps to try out something new.
- Detect the failure fast and further twist your initiatives.

6. Establish a Sustainable Environment and Work-Life Balance:

- Agile acknowledge the importance of a balanced and sustainable work environment.
- In a virtual context defining working hours and discouraging overworking contribute to a healthier work-life balance.

7. Visualize Everything:

- Visualization is crucial for planning and collaboration.
- In virtual collaboration, tools that provide visual representations of workflows, task assignments, and progress help teams stay organized, aligned, and responsive to changing needs.

Scenario based prototyping

Scenario-based prototyping is an approach in design thinking that involves creating realistic, detailed, and concise sequences of events to illustrate the activities of users in a real-world setting.

Steps in Scenario-Based Prototyping:

Use Scenarios:

- Analyze real-world situations through use of scenarios, which depict current activities

Design Scenarios:

- Revise and use scenarios to create design scenarios that illustrate how a new technology or system might be used. These scenarios serve as a vision for the future.

App Development Scenario:

- Consider the example of developing a mobile app for iOS or Android.

Customer Feedback:

- Instead of presenting the original app, create photos or visual representations using tools like Photoshop.
- These visuals include real scenarios, such as push-button interactions or directing to test cases.

Iteration and Revisions:

- UI and UX experts play a crucial role in refining the visual elements.
- Multiple iterations and revisions take place based on feedback, ensuring a user-friendly and efficient design.

Rolling Out Beta Versions:

- It's common practice to roll out beta versions to gather additional user feedback and make further improvements.

Types of Prototyping:

Rapid Prototyping/Throwaway:

- Quick creation of a prototype to explore design concepts.
- It may not be a final product but helps in validating ideas.

Evolutionary Prototyping:

- Continuous refinement of the prototype based on feedback and evolving requirements.

Scenario Components:

Rich Stories of Interaction:

- Detailed narratives that capture user interactions with the software.

Media Used:

- Plain text, pictures, sketches, screenshots, etc., can be used to articulate the scenarios.

Storyboards:

- Visual representations of scenarios that provide a holistic view of user interactions and experiences.

Benefits of Scenario-Based Prototyping:

- **Detailed Design Consideration:**

- Encourages thinking about the design in detail, helping identify potential problems before implementation.

- **User-Centered Approach:**

- Focuses on user activities and experiences, ensuring the final product aligns with user needs.

- **Feedback Integration:**

- Facilitates the integration of user feedback into the design process through iterative revisions.